AI-Powered Cyber Defense and Security Analytics Platform for Intelligent Threat Detection and Risk Assessment

Om Ganesh Dendge
Department of MCA
Indira College of Engineering & Management, Pune
An Autonomous Institute affiliated to SPPU
SY MCA, Semester III
Roll No./PRN: SYMCA46 / MC25OB1046
Email: om.dendge@indiraicem.ac.in

***Abstract*—The increasing volume, diversity of cyberattacks have created a practical limitation for security systems that rely primarily on predefined signatures, rules, and previously observed indicators. Machine learning (ML) provides a complementary capability by learning statistical and behavioral patterns from network traffic and using those patterns to identify suspicious activity. However, network intrusion detection is not only a classification problem. Security analysts must also understand the relative severity of alerts, reduce false-positive workload, investigate attack characteristics, and obtain interpretable evidence for prioritization.**

This paper presents the research design of an AI-powered cyber defense and security analytics platform that integrates machine-learning-based threat detection, attack classification, risk assessment, threat-informed analysis, and interactive visualization. The proposed framework uses publicly available benchmark datasets, primarily CIC-IDS2017 and UNSW-NB15, together with a controlled experimental workflow for preprocessing, feature selection, model training, and comparative evaluation. Candidate models include classical machine-learning baselines and selected ensemble/deep-learning models. The platform further introduces a risk-scoring layer that converts model output and threat characteristics into analyst-oriented priority levels and maps selected detections to the MITRE ATT&CK knowledge base.

The paper defines the research questions, system architecture, data-processing pipeline, experimental protocol, evaluation metrics, security considerations, and implementation design using Python-based analytics, FastAPI services, PostgreSQL, React.js, and authentication mechanisms. Importantly, no experimental performance values are fabricated: numerical results, confusion matrices, and comparative rankings are reserved for insertion after the planned experiments are completed. The intended contribution is an integrated, measurable, and academically reproducible prototype that studies the relationship between detection performance and operational security analytics.

***Index Terms*—cybersecurity, intrusion detection system, machine learning, security analytics, threat detection, risk assessment, MITRE ATT&CK, explainability, CIC-IDS2017, UNSW-NB15**

I. Introduction

Cybersecurity monitoring increasingly operates in an environment where malicious activity can evolve faster than manually maintained signatures and static rules. Conventional intrusion detection mechanisms remain valuable for known patterns, but their dependence on predefined knowledge can create limitations when attackers modify behavior, when large quantities of events must be processed, or when a previously unseen pattern appears. Machine learning provides a complementary approach by learning relationships from historical traffic and applying those learned patterns to new observations.

The proposed research follows this problem from a broader security-operations perspective. The research proposal identifies three connected needs: accurate threat detection, useful prioritization of detected events, and an analytical interface through which security information can be interpreted by an analyst. Public benchmark datasets such as CIC-IDS2017 provide labeled network-flow data containing benign and malicious activities, while UNSW-NB15 provides a separate benchmark containing modern normal activities and multiple contemporary attack categories [1], [2]. These datasets make it possible to conduct repeatable experiments before considering controlled laboratory traffic.

The research is therefore not limited to asking whether an event is malicious. It asks whether machine-learning methods can support a workflow in which suspicious activity is detected, classified, assigned a risk level, contextualized using threat-informed information, and presented through an interactive security analytics interface. MITRE ATT&CK is used as a common language for describing adversary tactics and techniques and for organizing detection-oriented analysis [5], [6].

The contribution of this work is consequently positioned at the intersection of ML-based intrusion detection and practical security analytics. The central design principle is that a high-performing classifier is useful only when its output can be transformed into information that supports analysis and prioritization. The resulting prototype is intended as an academic research system, not as a claim of production-scale replacement for enterprise SIEM, EDR, or SOC infrastructure.

II. Problem Statement

Conventional cybersecurity systems often depend heavily on signatures, rules, and known indicators of compromise. Such mechanisms can detect well-characterized attacks effectively, but operational limitations become more visible when traffic is high-volume, attack behaviors change, or previously unseen patterns appear. At the same time, security analysts may receive large numbers of alerts without sufficient context to determine which events require immediate attention.

ML-based intrusion detection has been extensively studied as a method for discovering statistical or behavioral patterns in network traffic. Nevertheless, reported performance can be sensitive to feature quality, class imbalance, dataset construction, training–testing distribution, and generalization to other environments. A model can also produce high aggregate accuracy while remaining operationally problematic because a comparatively small false-positive rate may still generate a large number of alerts in high-volume environments. Prior work has additionally emphasized that ML-based detection should not be assumed to generalize automatically beyond the conditions represented in the training data [3], [4].

Accordingly, the research problem addressed in this paper is:

> *How can an integrated AI/ML-based security analytics platform detect and classify suspicious network activity while also assigning meaningful risk priorities and presenting threat-oriented security insights in a form that is useful for analysts?*

III. Aim and Objectives

The aim is to design, implement, and experimentally evaluate an AI-powered cyber defense and security analytics platform that uses machine learning to detect suspicious network activity, classify potential threats, assess risk, and present actionable security information.

The objectives are:

1) To study existing AI/ML-based intrusion detection and security analytics techniques and identify their limitations regarding false positives, generalization, and operational use.
2) To preprocess and analyze network-security datasets and identify features relevant to malicious-activity detection.
3) To develop and compare suitable ML models for binary and, where feasible, multiclass attack classification.
4) To design a risk-assessment mechanism that prioritizes detected events according to model confidence and threat characteristics.
5) To integrate threat-informed context using MITRE ATT&CK tactics and techniques where a defensible mapping can be established.
6) To develop an interactive dashboard for monitoring alerts, risk levels, attack categories, model statistics, and analytical trends.
7) To evaluate the proposed approach using accuracy, precision, recall, F1-score, false-positive rate, detection rate, ROC-AUC, and processing-time measures.

IV. Research Questions and Hypotheses

The study investigates the following research questions:

- **RQ1:** How effectively can selected machine-learning techniques detect and classify malicious network activities?
- **RQ2:** Which selected model provides the best balance between detection performance and false-positive rate on the chosen benchmark?
- **RQ3:** How effectively can risk scoring improve the prioritization of detected security events?
- **RQ4:** Does an integrated security analytics dashboard improve the interpretability and operational usability of ML-based detection results?
- **RQ5:** How well does the proposed detection approach generalize across different attack categories and, where evaluated, across datasets?

Alternative hypothesis (H_1): An AI/ML-based security analytics approach can effectively detect and classify malicious network activities while providing useful risk prioritization compared with a selected conventional baseline.

Null hypothesis (H_0): The proposed AI/ML-based approach does not provide a statistically significant improvement in threat detection and risk prioritization compared with the selected baseline.

V. Background and Related Work

A. ML-Based Intrusion Detection

Machine-learning-based intrusion detection systems generally learn a decision boundary or representation from traffic attributes and use it to discriminate between benign and malicious observations. Classical models such as decision trees, random forests, support vector machines, and gradient-boosted approaches are often attractive because they can be trained on tabular flow features and can provide strong baselines. Neural approaches can learn nonlinear representations and may be useful when relationships among features are sufficiently complex.

A key methodological issue is evaluation outside the closed world represented by a dataset. Sommer and Paxson argued that intrusion-detection evaluation should be treated as a machine-learning problem embedded in an adversarial and changing environment, rather than as a conventional static classification task [3]. This concern motivates the use of multiple datasets or carefully controlled cross-category evaluation in the proposed study.

B. CIC-IDS2017

CIC-IDS2017 was designed to provide labeled benign traffic together with multiple contemporary attack scenarios and corresponding network-flow information [1]. The dataset can therefore support both binary detection and attack-type classification experiments. Its availability in flow-oriented CSV form is also useful for reproducible preprocessing and comparative tabular ML experiments [8].

C. UNSW-NB15

UNSW-NB15 was created from a mixture of modern normal activities and synthetic contemporary attack behaviors. The dataset contains multiple attack categories and generated features and is accompanied by predefined training and testing partitions [2], [9]. The existence of a distinct benchmark from CIC-IDS2017 makes it particularly useful for investigating whether a model's observed behavior is tied too closely to one dataset.

D. Threat-Informed Security Analysis

MITRE ATT&CK provides a knowledge base of adversary tactics and techniques derived from real-world observations. Tactics describe the adversary's technical objective, while techniques describe how that objective is achieved [6]. In the proposed system, ATT&CK is not treated as a substitute for model inference. Instead, it is used as a threat-context layer that can help analysts interpret an alert in terms of possible adversary behavior.

E. Robustness and Adversarial Machine Learning

Security-sensitive ML systems may themselves be exposed to adversarial manipulation, including attacks targeting data, models, or inference behavior. NIST's adversarial machine learning taxonomy emphasizes that AI systems introduce security considerations across the lifecycle and identifies attack goals and mitigation concepts that are relevant when evaluating trustworthy ML-based defenses [7]. This paper therefore treats robustness and generalization as limitations to be investigated rather than assuming that a high test accuracy is sufficient evidence of security.

F. Research Gap

The proposal underlying this paper identifies a practical gap between model-centric intrusion-detection research and integrated analyst-oriented security workflows. Many studies emphasize classification metrics for individual models, while practical monitoring requires additional functions such as alert prioritization, risk scoring, contextual interpretation, visualization, and reporting. The research gap addressed here is therefore the integration and experimental evaluation of these components in one coherent prototype.

VI. Proposed System

A. System Overview

The proposed platform is organized as a layered pipeline:

1) Data ingestion from benchmark datasets or controlled security-event sources.
2) Data quality checks, cleaning, encoding, normalization, and feature preparation.
3) ML inference for malicious-activity detection and attack classification.
4) Risk scoring and alert prioritization.
5) Threat-context enrichment using MITRE ATT&CK mappings where appropriate.
6) Persistence of alerts, predictions, scores, and metadata.
7) Interactive visualization and analytical reporting through a web dashboard.

B. Data Ingestion and Preprocessing

Network-flow records are first validated for schema consistency, missing values, duplicate observations, infinite values, and inconsistent labels. Categorical attributes are encoded using a consistent training-time mapping, while numerical variables can be standardized when required by the selected algorithm. Feature selection is performed using a combination of domain relevance and experimental evidence to reduce noise and computational cost.

To reduce information leakage, the train, validation, and test partitions are generated before any operation that learns distributional statistics from the data. Where stratified sampling is used, class proportions are preserved within the specified split. The final dataset counts and split ratios will be reported after preprocessing and quality checks.

C. Feature Representation

The primary research direction uses network-flow features rather than raw packet payloads. Candidate attributes may include duration, protocol, source/destination behavior, byte and packet counts, rate statistics, and other features supplied by the selected dataset. Features will be retained only when their semantics are documented and the preprocessing operation can be reproduced.

D. Comparative ML Models

The experimental design compares at least one classical baseline with one or more stronger candidate models. A practical baseline may be logistic regression or a decision tree, while tree ensembles such as random forest or gradient boosting can provide a more competitive tabular reference. If computationally feasible, a neural model can be included as an additional comparison.

The final paper should report the exact model versions, hyperparameters, random seeds, preprocessing steps, and training duration used in every experiment. This is essential for reproducibility and prevents performance claims from being detached from implementation details.

E. Risk Assessment Layer

Classification and confidence outputs are converted into a normalized risk score. A proposed prototype formulation is

$$R = 100\,(w_c C + w_s S + w_a A + w_m M), \tag{1}$$

where C represents calibrated model confidence, S represents estimated severity, A represents attack-category weighting, and M represents threat-context evidence. The weights satisfy

$$w_c + w_s + w_a + w_m = 1. \tag{2}$$

The variables and weights are design parameters, not empirical findings. They will be calibrated and sensitivity-tested during experimentation. Risk bands may be defined as

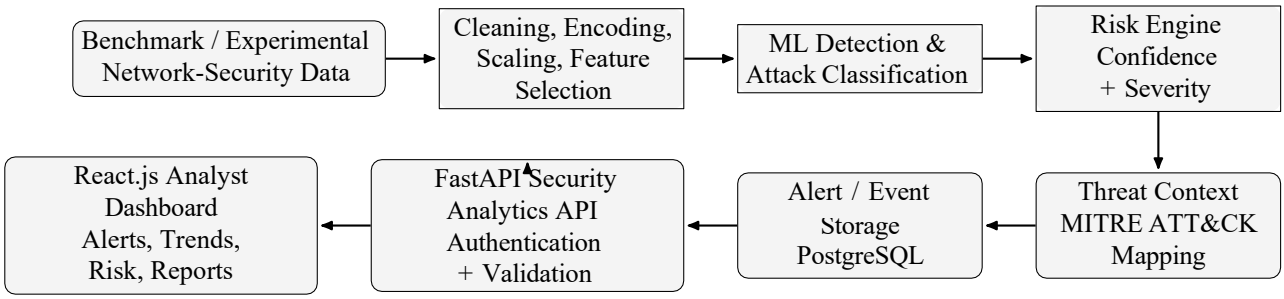


Fig. 1. Proposed end-to-end architecture of the AI-powered cyber defense and security analytics platform.

Low, Medium, High, and Critical using documented thresholds. The purpose of the score is prioritization rather than prediction of financial loss or exact incident impact.

F. MITRE ATT&CK Context

For attack classes with a defensible behavioral relationship to known ATT&CK techniques, the system can attach tactic and technique identifiers to an alert. This mapping should be represented explicitly as a rule-supported or analyst-reviewed enrichment layer. The system must not imply that a dataset label automatically proves a specific real-world ATT&CK technique.

G. Backend and Dashboard

The implementation stack follows the research proposal: Python-based analytics, FastAPI for service endpoints, PostgreSQL for structured storage, React.js for the dashboard, Pydantic for request and data validation, JWT for session/authentication tokens, and bcrypt for password-hash protection. FastAPI provides a typed Python web API framework with automatic interactive API documentation [12]. PostgreSQL provides structured relational storage and can also support JSON-oriented data when flexible alert metadata is required [13].

The dashboard is intended to expose:

- current and historical alert counts;
- risk distribution and high-priority events;
- attack-category statistics;
- model confidence and classification details;
- event-level analytical context;
- ATT&CK context where mapped;
- model-performance summaries and evaluation reports.

VII. Research Methodology

A. Research Design

The study is quantitative and experimental. It relies primarily on secondary data from public cybersecurity datasets and may include controlled experimental traffic for demonstration and validation. The research pipeline is designed so that model comparisons use the same source data, split protocol, feature pipeline, and test set.

TABLE I
Research Methodology Summary

Item	Planned Approach
Research type	Quantitative / Experimental
Primary data	CIC-IDS2017 and/or UNSW-NB15
Sample scale	At least 100,000 flow records where feasible
Sampling	Stratified dataset-based sampling
Preprocessing	Cleaning, encoding, scaling, feature selection
Models	Classical baseline + selected advanced models
Risk layer	Confidence + severity + category/context weighting
Threat context	MITRE ATT&CK enrichment where defensible
Analytics	Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Platform	FastAPI, React.js, PostgreSQL, JWT, bcrypt, Pydantic
Evaluation	Accuracy, precision, recall, F1, FPR, detection rate, ROC-AUC, time

B. Dataset Strategy

CIC-IDS2017 and UNSW-NB15 are the primary benchmark candidates. CIC-IDS2017 provides labeled network flows and packet captures for benign and multiple attack activities [8]. UNSW-NB15 contains 2,540,044 total records in four CSV files and also provides predefined training and testing subsets [9]. Because dataset size and class distribution can substantially influence training cost and evaluation, the final manuscript will document exactly which subsets are used.

A proposed experimental structure is:

- training set for fitting the model;
- validation set for model selection and threshold tuning;
- held-out test set for final evaluation.

Where possible, both within-dataset and cross-dataset experiments should be reported. The cross-dataset setting is particularly useful because a large performance reduction can reveal limited generalization.

C. Data Preprocessing

Let $\mathbf{x}$ denote a raw feature vector. Numerical normalization can be represented by

$$x_i' = \frac{x_i - \mu_i}{\sigma_i}, \tag{3}$$

where μ_i and σ_i are computed from the training set only. Categorical variables are encoded using the training mapping, and missing or invalid values are handled through a documented deterministic procedure.

D. Training and Validation

Each candidate model is trained using the same core training split. Hyperparameter optimization is restricted to training/validation data so that the held-out test set remains untouched until final evaluation. Random seeds are recorded when applicable. The resulting experiment should be reproducible through a configuration file or experiment manifest containing dataset version, preprocessing version, model configuration, random seed, and evaluation parameters.

VIII. Evaluation Framework

A. Classification Metrics

Let TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \tag{4}$$

$$Precision = \frac{TP}{TP + FP}, \tag{5}$$

$$Recall = \frac{TP}{TP + FN}, \tag{6}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}. \tag{7}$$

These measures are standard classification metrics and are available in the Scikit-learn evaluation framework [10].

For cybersecurity, recall is important because missed malicious events can be operationally expensive. Precision is also important because excessive false positives can increase analyst workload. Consequently, the paper will avoid selecting a model using accuracy alone.

The false-positive rate is defined as

$$FPR = \frac{FP}{FP + TN}. \tag{8}$$

Detection rate may be reported using recall for the malicious class, with the exact convention explicitly stated.

B. ROC-AUC

The receiver operating characteristic (ROC) curve describes the trade-off between true-positive rate and false-positive rate across decision thresholds. ROC-AUC provides a threshold-independent ranking summary and will be used alongside threshold-specific measures rather than as a single measure of security effectiveness [11].

C. Processing Time

The study will report at least inference latency per batch and, where feasible, records processed per second. These measures provide an indication of whether the proposed analytical workflow is computationally practical for an academic prototype.

D. Risk-Prioritization Evaluation

Risk scoring will be evaluated through:

1) distribution of alerts across risk bands;
2) concentration of known high-severity attack records within higher risk bands;
3) score stability under repeated evaluation;
4) sensitivity to changes in model confidence and severity weights.

The research will not claim that the risk formula predicts real incident impact unless empirical validation supports such a claim.

IX. Threat Modeling and Security Considerations

A. Defensive Scope

The platform is restricted to defensive activities: detection, classification, monitoring, analysis, risk prioritization, and reporting. It does not perform unauthorized attacks against external systems and is not intended to replace a full enterprise SIEM, EDR, or SOC platform.

B. Security of the Platform

The application itself is part of the attack surface. The implementation therefore includes authentication, authorization boundaries, input validation, secure password hashing, and database controls. API endpoints should validate structured input using Pydantic models and reject malformed or unexpected payloads. Sensitive secrets must be stored outside source code.

C. Adversarial Robustness

Because security models can be manipulated through adversarial data or distribution changes, the research will document this issue explicitly. NIST's taxonomy highlights threats such as evasion and poisoning and emphasizes lifecycle-level security considerations for ML systems [7]. The prototype will treat adversarial robustness as an evaluation limitation and a future experimental extension unless sufficient time and compute are available for dedicated robustness testing.

X. Experimental Procedure

The complete experiment is structured as follows:

1) Acquire and version the selected public datasets.
2) Inspect data quality, labels, class distributions, and duplicated or invalid records.
3) Create a leakage-resistant stratified train/validation/test split.
4) Apply preprocessing fitted only on training data.
5) Train the selected classical baseline.
6) Train the selected advanced model(s) under the same evaluation conditions.
7) Tune decision thresholds using validation data.
8) Run final inference on the held-out test set.

9) Calculate classification and efficiency metrics.
10) Generate confusion matrices and ROC curves.
11) Apply the risk-scoring layer to the final predictions.
12) Enrich selected events with MITRE ATT&CK context.
13) Persist results in the analytics platform and expose them through the dashboard.
14) Perform statistical and qualitative comparison and document limitations.

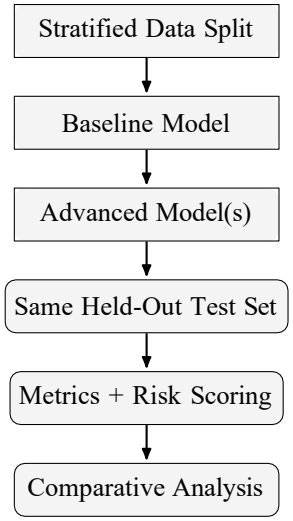


Fig. 2. Controlled comparative evaluation procedure.

XI. Results and Discussion Framework

At the current stage, the uploaded project proposal defines the research problem, methodology, expected outcomes, and evaluation plan but does not provide completed experimental measurements. Consequently, numerical values in this draft are intentionally not fabricated. The following tables are designed to be populated after implementation and testing.

TABLE II
Final Model Performance Comparison Template

Model	Acc.	Prec.	Rec.	F1	FPR
Baseline	TBD	TBD	TBD	TBD	TBD
Model 2	TBD	TBD	TBD	TBD	TBD
Model 3	TBD	TBD	TBD	TBD	TBD

TABLE III
Operational Evaluation Template

Model	ROC-AUC	Time/Batch	Risk Ranking
Baseline	TBD	TBD	TBD
Model 2	TBD	TBD	TBD
Model 3	TBD	TBD	TBD

Once the experiments are complete, the discussion should answer each research question directly. The preferred structure is to first report the measured result, then interpret the finding, then compare it with the expected behavior from the literature, and finally discuss operational implications and limitations. This avoids replacing research evidence with assumptions.

A. Confusion Matrix Analysis

The final paper should present a confusion matrix for the best-performing candidate and, where space permits, for all major models. False positives should be analyzed separately because they affect alert volume and analyst workload. False negatives should also receive explicit attention because missed malicious events are security-critical.

B. Generalization Analysis

If both CIC-IDS2017 and UNSW-NB15 are used, the final discussion should report whether performance remains stable across datasets. A substantial drop in performance should not be described simply as model failure; it may provide evidence that the learned representation is dataset-specific. Such a finding would itself be relevant to RQ5.

C. Risk Analytics Analysis

The research should analyze whether higher-risk alerts concentrate events with high attack severity or high-confidence malicious predictions. The analysis should distinguish between a score that is mathematically ordered and a score that is empirically validated. A risk ranking can be operationally useful without being interpreted as a ground-truth estimate of business impact.

XII. Expected Contributions

The intended contributions of this research are:

1) an integrated prototype that combines ML-based network threat detection with security analytics;
2) a reproducible preprocessing and evaluation pipeline based on public benchmark datasets;
3) an experimental comparison of selected ML models using security-relevant metrics;
4) a risk-prioritization mechanism that transforms detection outputs into analyst-oriented alert levels;
5) a threat-context layer based on MITRE ATT&CK mappings where justified;
6) an interactive dashboard for visualization, analysis, and reporting;
7) a documented analysis of dataset limitations, false positives, generalization, and computational cost.

XIII. Limitations

Several limitations are recognized in advance. First, public benchmark datasets do not reproduce the complete diversity of live enterprise network environments. Second, differences in labeling, traffic generation, feature engineering, and class balance can influence reported performance. Third, a model that performs strongly on one benchmark may not generalize to previously unseen attack behavior. Fourth, the proposed risk score is an operational prioritization mechanism and requires validation before it can be treated as a reliable measure

of real-world incident impact. Finally, the platform is an academic prototype and therefore does not provide the breadth, redundancy, governance, integrations, and scale of a production SOC or SIEM.

XIV. Ethical and Responsible Use

The project is defensive in scope. Only authorized datasets, controlled laboratory traffic, and permitted security environments should be used. No external system should be scanned, attacked, or monitored without authorization. If real organizational logs are ever introduced in future work, access control, data minimization, anonymization, retention policies, and institutional approval should be considered.

The study also follows a responsible reporting principle: experimental values must be derived from actual runs, scripts, and stored artifacts. Results should not be selected selectively to support the hypothesis, and failures or unexpected observations should be reported when they materially affect the conclusions.

XV. Implementation and Reproducibility Plan

To make the research suitable for technical review, the final implementation should maintain a reproducible project structure with separate components for data preparation, feature engineering, model training, evaluation, backend services, and dashboard presentation. Every experiment should produce machine-readable outputs such as metrics JSON/CSV, confusion-matrix data, ROC data, model configuration, and execution metadata.

A minimal reproducibility record should contain:

- dataset source and version/date accessed;
- selected records and class counts;
- train/validation/test split strategy;
- feature list and preprocessing configuration;
- model algorithm and hyperparameters;
- software library versions;
- hardware profile and training duration;
- random seeds;
- final evaluation metrics;
- known limitations and failed experiments.

XVI. Work Plan

The project follows the proposal timeline of approximately 17–18 weeks:

TABLE IV
Research Work Plan

Phase	Duration
Literature review	2 weeks
Requirement analysis and design	2 weeks
Dataset preparation	2 weeks
ML model development	3 weeks
Platform implementation	4 weeks
Testing and evaluation	2 weeks
Documentation and final presentation	2 weeks

XVII. Conclusion

This paper has presented a complete research framework for an AI-powered cyber defense and security analytics platform focused on intelligent threat detection and risk assessment. The study moves beyond a narrow classification objective by integrating data preparation, machine-learning detection, risk prioritization, threat-informed context, persistent alert storage, and analyst-facing visualization.

The research is grounded in publicly available cybersecurity benchmarks and a controlled experimental methodology. CIC-IDS2017 and UNSW-NB15 provide complementary benchmark settings, while the evaluation framework emphasizes precision, recall, F1-score, false-positive rate, ROC-AUC, and processing time rather than accuracy alone. MITRE ATT&CK is positioned as a contextual enrichment layer rather than as a substitute for empirical ML evidence.

At the present stage, the work is a research design and implementation blueprint rather than a completed experimental study. The final numerical results, charts, confusion matrices, and model selection should be inserted only after the experiments are executed. This distinction is important for academic integrity and for ensuring that the final conclusions are supported by evidence. The completed study is expected to provide practical insight into whether integrated ML-based security analytics can improve not only threat detection, but also the prioritization and interpretation of cybersecurity events.

References

[1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in *Proc. 4th International Conference on Information Systems Security and Privacy (ICISSP)*, 2018.

[2] N. Moustafa and J. Slay, "UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems (UNSW-NB15 Network Data Set)," in *Proc. Military Communications and Information Systems Conference (MilCIS)*, 2015.

[3] R. Sommer and V. Paxson, "Outside the Closed World: On Using Machine Learning for Network Intrusion Detection," in *Proc. IEEE Symposium on Security and Privacy*, 2010, pp. 305–316.

[4] Y. Guo, "A Review of Machine Learning-Based Zero-Day Attack Detection: Challenges and Future Directions," *Computer Communications*, vol. 198, 2023.

[5] B. E. Strom *et al.*, "Finding Cyber Threats with ATT&CK-Based Analytics," MITRE, 2017.

[6] MITRE, "MITRE ATT&CK: Globally Accessible Knowledge Base of Adversary Tactics and Techniques," MITRE Corporation. [Online]. Available: https://attack.mitre.org/

[7] A. Vassilev, A. Oprea, A. Fordyce, H. Anderson, X. Davies, and M. Hamin, "Adversarial Machine Learning: A Taxonomy and Terminology of Attacks and Mitigations," NIST AI 100-2e2025, National Institute of Standards and Technology, 2025.

[8] Canadian Institute for Cybersecurity, "CICIDS2017 Dataset." [Online]. Available: https://www.unb.ca/cic/datasets/ids-2017.html

[9] UNSW Canberra Cyber, "The UNSW-NB15 Dataset." [Online]. Available: https://research.unsw.edu.au/projects/unsw-nb15-dataset

[10] Scikit-learn Developers, "sklearn.metrics: Score Functions, Performance Metrics and Pairwise Metrics." [Online]. Available: https://scikit-learn.org/stable/api/sklearn.metrics.html

[11] Scikit-learn Developers, "roc_auc_score." [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_auc_score.html

[12] FastAPI, "FastAPI Documentation." [Online]. Available: https://fastapi.tiangolo.com/

[13] PostgreSQL Global Development Group, “JSON Types – PostgreSQL Documentation.” [Online]. Available: https://www.postgresql.org/docs/17/datatype-json.html